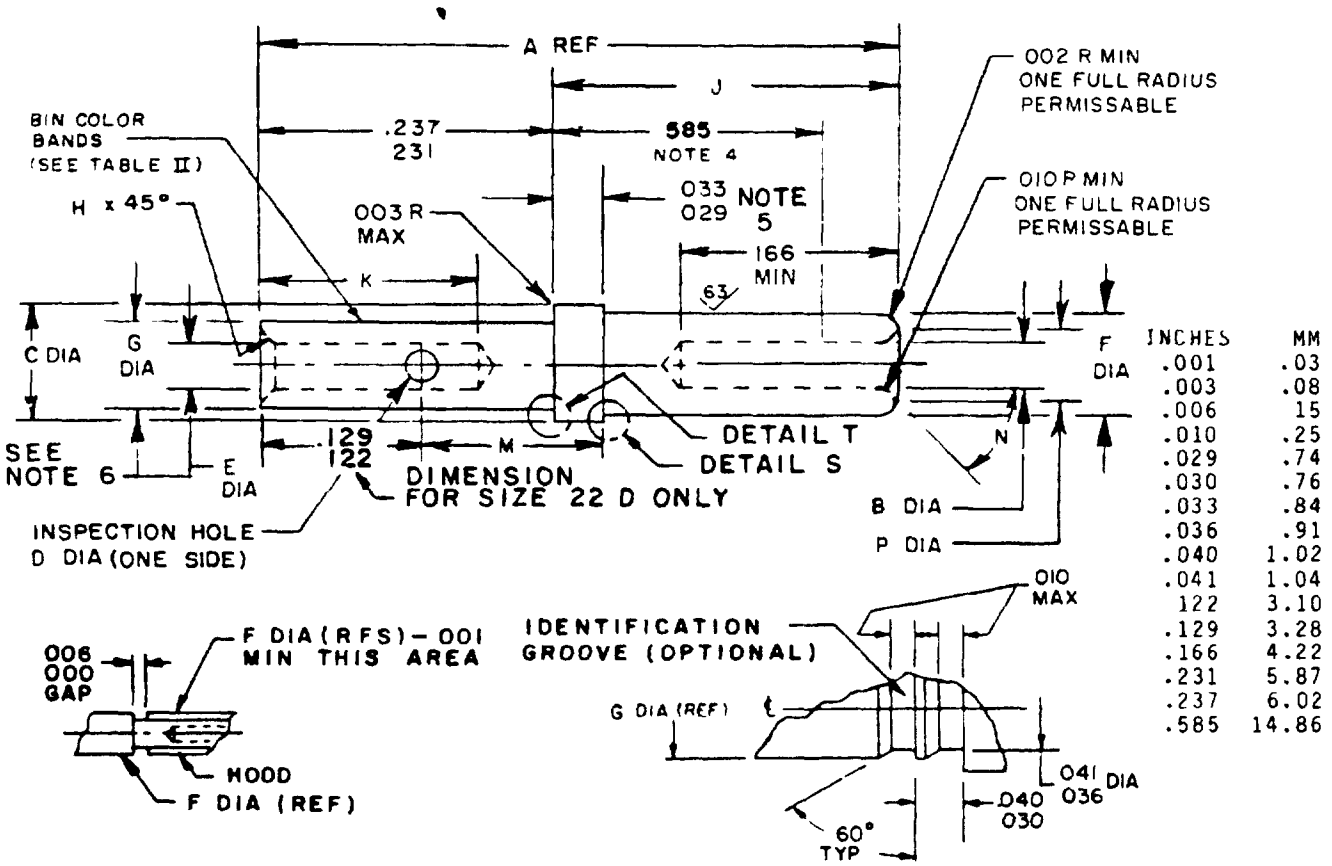
MILITARY SPECIFICATION SHEET

CONTACTS, ELECTRICAL CONNECTOR, SOCKET, CRIMP REMOVABLE, THERMOCOUPLE (FOR MIL-C-38999 SERIES I, II, AND IV CONNECTORS)

This specification is approved for use by all Departments and Agencies of the Department of Defense

The complete requirements for acquiring the contacts described herein shall consist of this specification and the latest issue of MIL-C-39029.



INCHES	MM
.001	.03
.003	.08
.006	15
.010	.25
.029	.74
.030	.76
.033	.84
.036	.91
.040	1.02
.041	1.04
122	3.10
.129	3.28
.166	4.22
.231	5.87
.237	6.02
.585	14.86

DETAIL S (ALL SIZES)

DETAIL T SIZE 22 D ONLY

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm
3. Dimensions shown apply after plating
4. Point at which a square ended pin of the same basic diameter as the mating contact first engages the socket contact spring
5. Indicated dimension does not apply for size 22D.
6. For size 22D only, diameters F and G to be concentric within .003 (TIR) regardless of feature size (RFS), for all other contact sizes, diameters E and G to be concentric within .001 (TIR) at maximum material condition (MMC).
7. Hoods shall not exceed contact body diameter regardless of feature size (RFS) in attachment area

FIGURE 1. Socket contacts

FSC 5935

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INCHES	MM	INCHES	MM	INCHES	MM
.002	.05	.044	1.07	.094	2.39
.003	.08	.046	1.17	.101	2.57
.005	.13	.047	1.19	.103	2.62
.010	.25	.048	1.22	.113	2.87
.018	.46	.053	1.35	.127	3.23
.020	.51	.060	1.52	.130	3.30
.022	.56	.062	1.57	.141	3.58
.026	.66	.064	1.63	.209	5.31
.029	.74	.066	1.68	.229	5.82
.031	.79	.068	1.73	.231	5.87
.032	.81	.070	1.78	.237	6.02
.033	.84	.072	1.83	.576	14.63
.0335	.85	.078	1.98	.616	15.65
.0355	.90	.082	2.08	.626	15.90
.036	.91	.084	2.13	.855	21.72
.0366	.93	.088	2.24		
.042	1.07	.091	2.31		

TABLE I Dimensions.

BIN code	A DIA	B DIA	C DIA	D DIA	E DIA	F DIA MAX	G	H	J	K	M	N°	P DIA MIN
482 through 485	.855	.031	.062	.022	.0355	.062	.048	.005	.626	.141	---	50°	.047
		(MIN)	.060	.018	.0335		.046	.003	.616	(MIN)	---	44°	
486 through 489		.044	.094	.032	.048	.078	.070	.010		.229	.078	47°	053
		.042	091	.026	.046		.068	.005		.209	.072	40°	
490 through 493		.064	.130	.042	.068	113	.103			229	.088	47°	.084
		(MIN)	.127	.036	.066		.101			209	.082	40°	

FIGURE 1. Socket contacts - Continued.

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TABLE II. Design characteristics.

code	Color bands			Mating end size	Wire barrel size	Type	Class
	1st.	2nd.	3rd.				
82	Yellow	Gray	Red	22	22D	C (See table IV)	B
83	Yellow	Gray	Orange	22	22D		
84	Yellow	Gray	Yellow	22	22D		
85	Yellow	Gray	Green	22	22D		
86	Yellow	Gray	Blue	20	20		
87	Yellow	Gray	Violet	20	20		
88	Yellow	Gray	Gray	20	20		
89	Yellow	Gray	White	20	20		
90	Yellow	White	Black	16	16		
91	Yellow	White	Brown	16	16		
92	Yellow	White	Red	16	16		
93	Yellow	White	Orange	16	16		

TABLE III. Tools.

N ode	Basic crimping tool	Positioner	Installing tool	Removal tool
12	M22520/2-01	M22520/2-07	M81969/14-01	M81969/14-01
13	M22520/7-01	M22520/7-05	M81969/8-01	M81969/8-02
14				
15				
16	M22520/1-01	M22520/1-04	M81969/8-05	M81969/8-06
17	M22520/2-01	M22520/2-10	M81969/14-02	M81969/14-02
18	M22529/7-01	M22520/7-08		
19				
91	M22520/1-01	M22520/1-04	M81969/8-07	M81969/8-08
91	M22520/7-01	M22520/7-04	M81969/14-03	M81969/14-03
92				
93				

TABLE IV. Materials, plating, and tensile strength.

BIN code	Material	Plating	Tensile strength 1/			
			Wire size	Axial load	Wire size	Axial load
				lbs		lbs
482	Constantan	None	28	2.25	22	7.5
483	Alumel	None				
484	Chromel	None				
485	Iron	Cadmium Plate 2/				
486	Constantan	None	24	6.00	20	12.0
487	Alumel	None				
488	Chromel	None				
489	Iron	Cadmium Plate 2/				
490	Constantan	None	20	14.00	16	45.0
491	Alumel	None				
492	Chromel	None				
493	Iron	Cadmium plate 2/				

pplies to type 1 wire per MIL-W-5846
romate clear coat.

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REQUIREMENTS

Dimensions, design characteristics, and configuration See figure 1 and tables I and II

Mating contacts MIL-C-39029/87.

Tools See table III.

Materials and plating See table IV.

Tensile strength See table IV

Vibration Test condition VI, letter J of method 2005 of MIL-STD-1344, except duration shall be 8 hours in the longitudinal direction and 8 hours in the perpendicular direction (16 hours total).

Shock (specified pulse) Test condition D of method 2004 of MIL-STD-1344, except the pulse shall be an approximate half sine wave of 300 g ±15 percent magnitude with a duration of 3 milliseconds ±1.

QPL evaluating activity Defense Electronics Supply Center (DESC-E), Dayton, Ohio, 45444.

International interest NEPR 57.

Part numbers See table V.

TABLE V. Part number and BIN code.

BIN code	Part number	Supersedes
482	M39029/88-482	M39029/882222C1
483	M39029/88-483	M39029/882222C2
484	M39029/88-484	M39029/882222C3
485	M39029/88-485	M39029/882222C4
486	M39029/88-486	M39029/882020C1
487	M39029/88-487	M39029/882020C2
488	M39029/88-488	M39029/882020C3
489	M39029/88-489	M39029/882020C4
490	M39029/88-490	M39029/881616C1
491	M39029/88-491	M39029/881616C2
492	M39029/88-492	M39029/881616C3
493	M39029/88-493	M39029/881616C4

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians
Army - CR
Navy - AS
Air Force - 85

Review activities
Army - AR
Navy - EC
Air Force - 11, 17, 99
DLA - ES

User activities
Navy - OS, SH

Agent
DLA - ES

Preparing activity
Air Force - 85

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